

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title: COMPLETED RESEARCH		Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 12

(1) Research Program/Project/Study Title: <i>Pest and Soil Fertility Dynamics Under Organic Cultivation System</i>	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center							
(4) Site of implementation <table border="1" style="width: 100%;"> <tr> <td>Municipality/City:</td> <td>Mambusao</td> </tr> <tr> <td>Province:</td> <td>Capiz</td> </tr> <tr> <td>Region:</td> <td>VI</td> </tr> </table>				Municipality/City:	Mambusao	Province:	Capiz	Region:	VI
Municipality/City:	Mambusao								
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(5) Classification <table border="1" style="width: 100%;"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Vegetable		(8) Discipline/Field of Specialization Agriculture							

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 12

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Organic Agriculture Project of Capiz State University, Burias Campus, Mambusao, Capiz. Data collection encompassed the quantification of non-beneficial and beneficial pests, and soil analysis, including pH, potassium (K), phosphorus (P), and nitrogen in organic matter (OM). Additionally, vegetable yield was measured in terms of tons per hectare (t/ha). The selected vegetables for this study comprised tomato, radish, string beans, eggplant, ampalaya, pepper, sweetcorn, and kangkong. This was conducted to determine the dynamics of pests and soil fertility under organic cultivation.

The results showed that whitefly was the most prevalent pest (3,020), followed by the green leaf hopper (2,546) and, lastly, the green lopper (1,750). Finally, a positive and steady trend in soil fertility dynamics was revealed in terms of pH ($6.79 < 7.3 < 6.77$), % organic matter ($1.88 < 2.49 < 2.92$), exchangeable potassium ($249.32 < 315.96 < 367.95$), and available phosphorus (P) ($16.89 > 11.14 < 56.74$).

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

organic farming, organic culture, organic vegetable production, sustainability, vegetables, beneficial insects

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Organic cultivation practices have gained significant attention in recent years, driven by growing concerns about the environmental impact of conventional agriculture, pesticide residues, and the long-term sustainability of food production (Reganold and Wachter, 2016). Organic farming emphasizes the use of natural and sustainable techniques to manage pests and enhance soil fertility. Understanding the intricate relationship between pests and soil fertility dynamics in organic agriculture is essential for optimizing crop productivity and sustainability.

Pest management is a critical component of organic cultivation, as synthetic chemical pesticides are generally prohibited in organic systems (Reed, 2017). Organic growers rely on a combination of strategies, such as biological control, cultural practices, and the use of organic pesticides, to manage pests and diseases. The effectiveness of these strategies can be influenced by factors like crop diversity, crop rotation, and cover cropping, which also play a role in maintaining and enhancing soil fertility (Mäder et al., 2002). Therefore, a holistic approach to pest management in organic systems must consider the dynamic interactions with soil fertility.

Soil fertility is a central factor that directly impacts crop health and productivity in organic agriculture (Gosling et al., 2006). Organic practices, including the use of compost, cover crops, and reduced tillage, aim to enhance soil fertility by increasing organic matter, nutrient availability, and beneficial soil microorganisms. The relationships between soil fertility dynamics and pest management are complex and

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 12

multifaceted. For example, the presence of beneficial soil organisms can help control pest populations, while the availability of specific nutrients may influence plant resistance to pests (Birkhofer et al., 2017).

This study seeks to explore the intricate connections between pest management and soil fertility dynamics in organic farming systems. By examining the interplay between pest pressures, soil fertility parameters, and various pest management strategies within the context of organic agriculture, this research contributes to a deeper understanding of how sustainable farming practices can be optimized to balance the needs of crop protection and soil health.

Objectives of the Study

In general, the purpose of the study was to determine the pest and soil fertility dynamics under organic cultivation. Specifically, the study aimed to:

1. To ascertain the pest occurrence in an organic cultivation environment;
2. To investigate the changes in soil fertility dynamics in organic agricultural systems grown with vegetables; and
3. To systematically document and assess the trends in pest occurrence, and soil fertility dynamics of an organic cultivation system.

(11) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Experimental Method

The study was conducted at the Organic Agriculture Project of Capiz State University, Burias Campus, Mambusao, Capiz. Data collection encompassed the quantification of non-beneficial and beneficial pests, and soil analysis, including pH, potassium (K), phosphorus (P), and nitrogen in organic matter (OM).

Cultural Practices and Management

Pest Management

Pest populations were determined through meticulous visual observation and recorded on a monthly basis.

Soil Fertility Dynamics

Before commencing the experiment, soil samples were collected to evaluate the chemical properties of the soil. Composite soil samples were air-dried, sieved through a 2-mm mesh, and sent to the Regional Soils Laboratory for analysis. Soil sample collection was performed to determine pH (potentiometric method), % organic matter (S. A. Wides method), exchangeable potassium (K), and available phosphorus (Olsen's P method).

(12) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 12

figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Prevalence of pest

Figure 1 shows the data gathered on non-beneficial pests in year 1 (2021). Data revealed that prevalent pests were whiteflies (3,020); green loppers (1,750); orange beetles (1,540); and flies (230). Furthermore, Figure 2 shows the data gathered in year 2 (2022). Data revealed that prevalent pests were rice bugs (1,591); green leaf hoppers (1,475); orange beetles (1,415); thrips (689); DBM (486); grasshoppers (472); leaf folder (309); semi-loppers (282); flies (265); moths (183); whorl maggots (159); steam borders (39); corn borders (33); fruit borders (25); and cutworms (12). Finally, Figure 3 shows the data gathered in year 3 (2023). Data revealed that prevalent pests were green leaf hoppers (2,546); leaf folders (1,358); orange beetles (1,240); DBMs (410); flies (380); thrips (113); and moths (82).

On the other hand, Figure 4 revealed that the prevalence of beneficial pests in year 1 (2021) were dragonflies (6); ladybugs (4); and spiders (2). Figure 5 revealed that in year 2 (2022) the pests found were ladybugs (67); dragonflies (60); and spiders (31). Figure 6 revealed that in year 3 (2023) the pest were spiders (11); ladybugs (3); and dragonflies (3).

Based on the results, the highest number of non-beneficial pests in three consecutive years were whitefly, rice bug, and green leaf hopper, respectively. While the dragonfly, ladybug, and spider got the highest number of beneficial pests in the same consecutive years.

After visual observation and manual counting, a consistent increase in the total pest population affecting vegetables throughout the years was detected. A combined total of 21,811 non-beneficial (Figure 7) and 101 beneficial (Figure 8) pests were identified in Year 1, Year 2, and Year 3.

These findings substantiate the research conducted by De Ponti et al. (2012), which underscores the prevalence of pests, diseases, and other yield-reducing factors, particularly those associated with nutrient availability, in organic agriculture. However, the adaptive characteristics of certain crop varieties to their environment have equipped them with more efficient pest management mechanisms, consequently lowering the risk of infection and pest infestation (IFOAM, 2016).

Soil Analysis Trend

A noticeable trend can be observed in the pH content analysis from Year 1 to Year 3. Soil pH is a critical parameter that exerts a significant influence on various soil physical, chemical, and biological processes and properties. In particular, the pH results as shown in Figure 9 revealed a value of 6.79 in Year 1, with a slight increase in Year 2 (7.3), followed by a marginal decrease in Year 3 (6.77).

The results concerning the organic matter content Nitrogen (N (OM)) as

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 12

reflected in Figure 10 indicated a consistent increase in soil organic matter levels. The trend showed that in Year 1, the N (OM) content was 1.48, followed by incremental increases in the subsequent years (Year 2 = 2.49; Year 3 = 2.92). This finding aligns with the study conducted by Dhaliwal et al. (2019), which emphasized that organic matter accumulation enhances the dynamics of soil micronutrient fractions.

Conversely, the analysis of potassium (K) content as shown in Figure 11 displayed a steady increase over the years. Year 1 recorded a value of 249.32, which was followed by 315.96 in Year 2, with a further incremental increase in Year 3 (367.95). Furthermore, the trend in soil phosphorus (P) analysis revealed a value of 16.89 in Year 1, with a minor decrease in Year 2 (11.14) and a substantial increase in Year 3 (56.74). This underscores the importance of phosphorus as a major macronutrient, acting as an energy activator that significantly influences crop yield and quality, as noted by Toledo et al. (2021).

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the study's findings, the following conclusions have been reached:

1. A total of 21,710 non-beneficial pests and 101 beneficial pests were identified. The most prevalent among them were whiteflies, green leaf hoppers, and green loppers.
2. There was a fluctuating trend on the number of beneficial insects from year 1 to year 3.
3. The soil analysis trends indicate a consistent increase in pH levels, nitrogen (N) content, and potassium (K) content. However, phosphorus (P) analysis showed fluctuations in activity.

Recommendations

Based on the findings of this study, the following recommendations are forwarded:

1. Adapt organic crop production system to increase the number of beneficial insects such as spider;
2. Implement organic cultivation to increase soil pH, nitrogen and potassium content.
3. Implement cultural practices for organic vegetable cultivation and maintain regular monitoring to control prevalent non-beneficial pests such as whiteflies and green leaf hoppers; and encourage natural predators, track pest and predator data, and make informed decisions for effective pest management in organic vegetable farming.
4. Recognize distinct nutrient requirements of different vegetable crops and use

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 12

crop rotation to balance soil nutrients and reduce reliance on synthetic fertilizers; and select crop rotations that consider soil nutrient needs and the previous year's crop to maintain soil health and support sustainable farming practices.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

Birkhofer, K., Bezemer, T. M., Bloem, J., Bonkowski, M., Christensen, S., Dubois, D., ... & Geisen, S. (2017). Long-term organic farming fosters below and aboveground biota: implications for soil quality, biological control and productivity. *Soil Biology and Biochemistry*, 112, 44-57.

De Ponti, T., Rijk, B., & Van Ittersum, M. (2012). The crop yield gap between organic and conventional agriculture. *Agricultural Systems*, 108, 1–9. <https://doi.org/10.1016/j.agsy.2011.12.004>

Dhaliwal, S. S., Naresh, R., Mandal, A., Singh, R., & Dhaliwal, M. (2019). Dynamics and transformations of micronutrients in agricultural soils as influenced by organic matter build-up: A review. *Environmental and Sustainability Indicators*, 1–2, 100007. <https://doi.org/10.1016/j.indic.2019.100007>

Gosling, P., Shepherd, M., & Davies, M. (2006). Beech strip intercropping for increased resource use efficiency and ecological enhancement of hill pastures. *Agriculture, Ecosystems & Environment*, 117(3-4), 227-237.

Mäder, P., Fließbach, A., Dubois, D., Gunst, L., Fried, P., & Niggli, U. (2002). Soil fertility and biodiversity in organic farming. *Science*, 296(5573), 1694-1697.

Reed, M. (2017). Integrated Pest Management in Organic Systems. In *Organic Farming for Sustainable Agriculture* (pp. 267-289). Elsevier.

Reganold, J. P., & Wachter, J. M. (2016). Organic agriculture in the twenty-first century. *Nature Plants*, 2(2), 15221.

Toledo, R. L., Reyes-Martín, M. P., Celi, L., & Ondoño, E. F. (2021). Phosphorus Dynamics in the Soil–Plant–Environment Relationship in Cropping Systems: a

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 12

review. Applied Sciences, 11(23), 11133. <https://doi.org/10.3390/app112311133>

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 1 of 12

APPENDICES

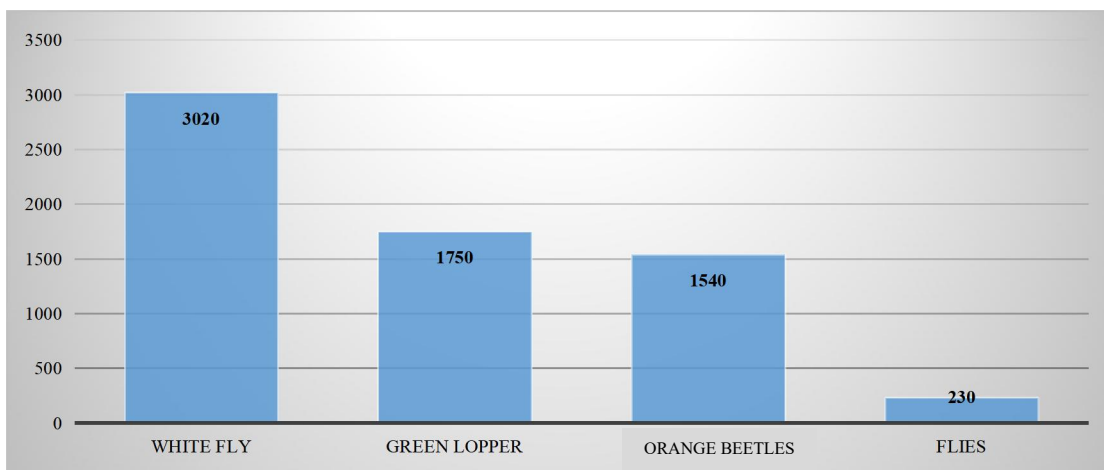


Figure 1. A graphic chart of non-beneficial pests in Year 1 (2021).

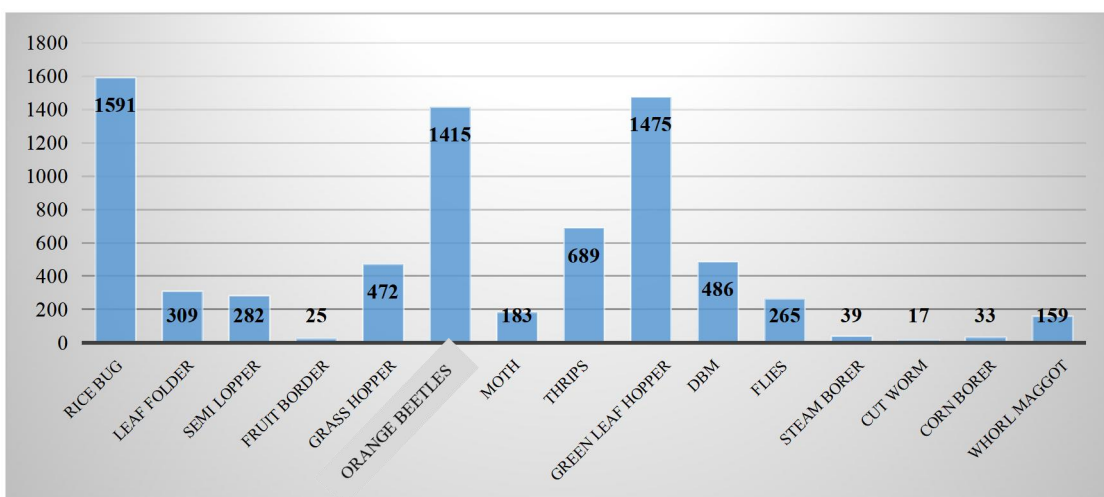


Figure 2. A graphic chart of non-beneficial pests in Year 2 (2022).

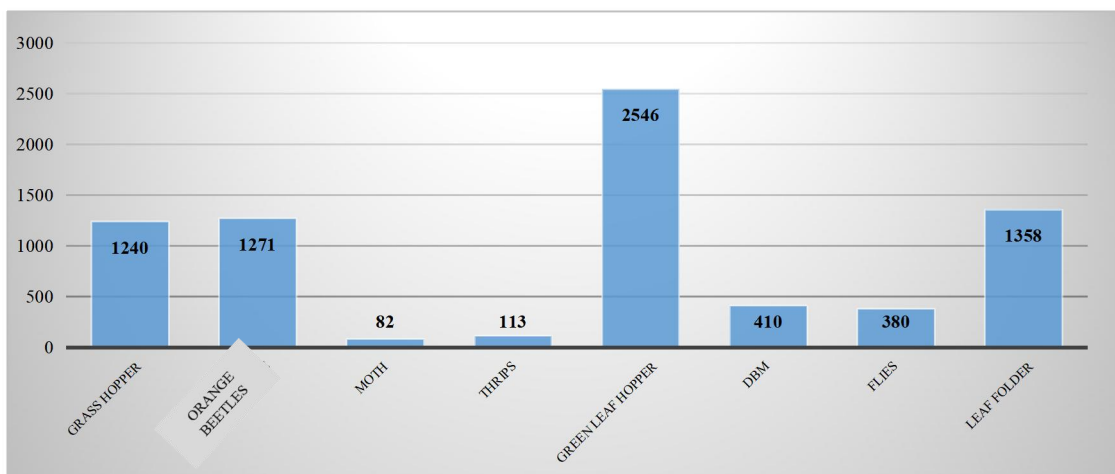


Figure 3. A graphic chart of non-beneficial pests in Year 3 (2023).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title: COMPLETED RESEARCH		Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 2 of 12

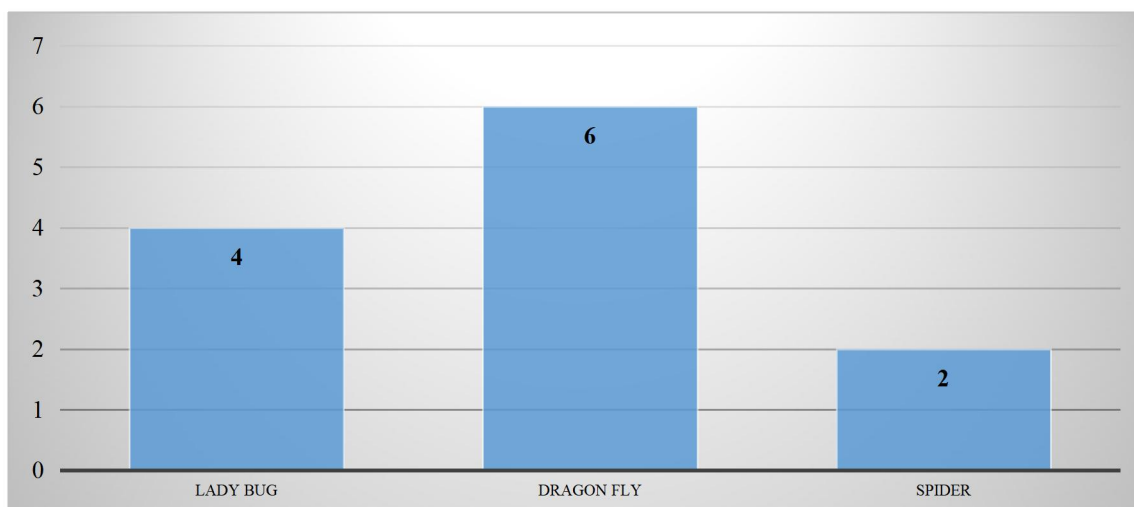


Figure 4. A graphic chart of beneficial pests in Year 1 (2021).

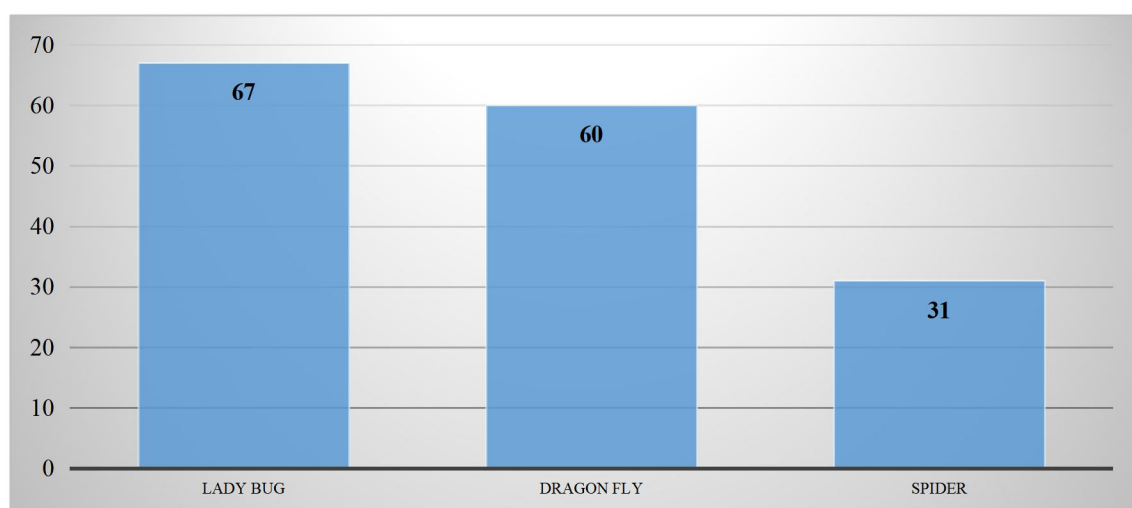


Figure 5. A graphic chart of beneficial pests in Year 2 (2022).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 12

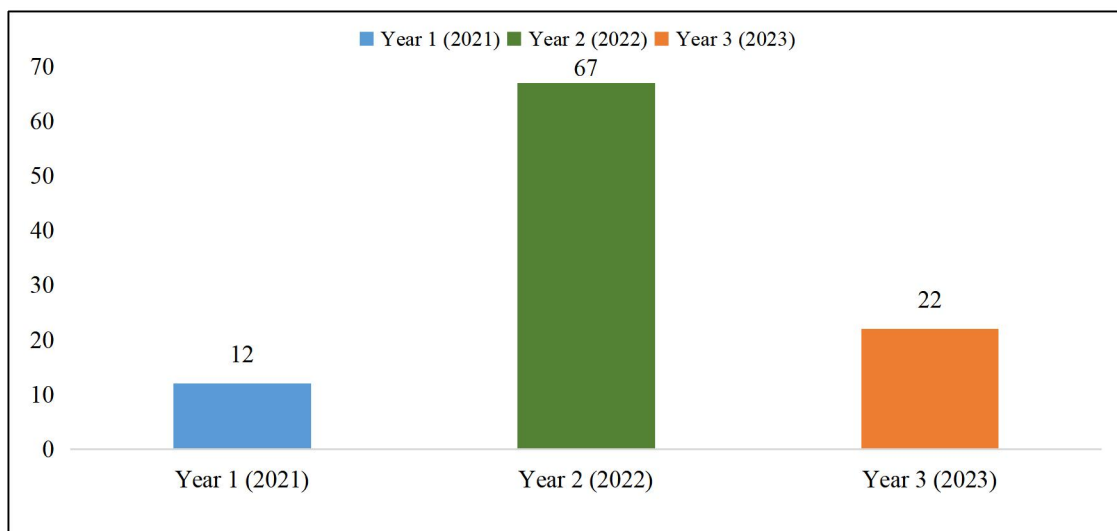


Figure 7. A graphic chart of the total non-beneficial pests per year.

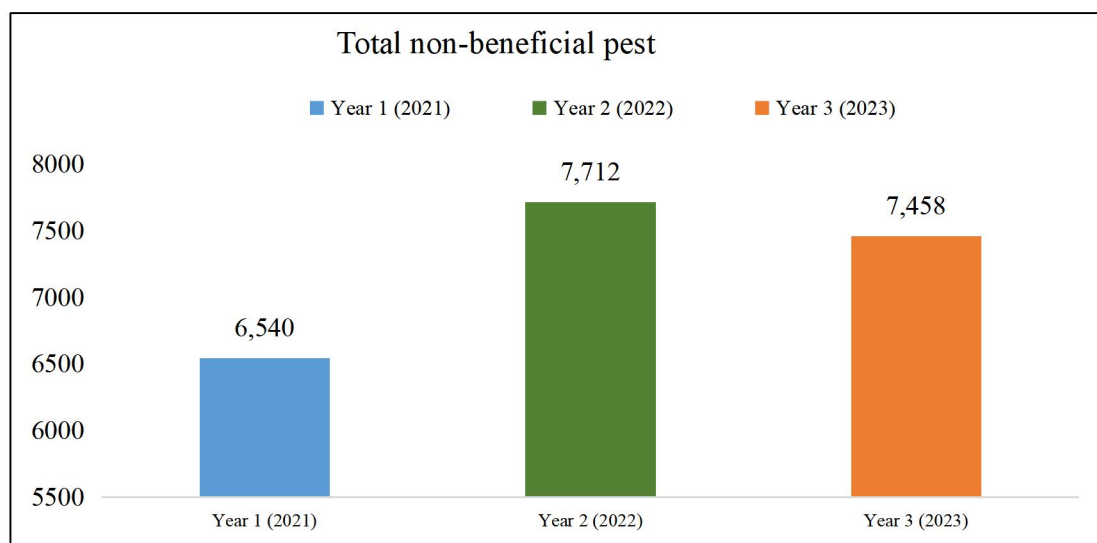


Figure 8. A graphic chart of the total beneficial pests per year.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 12

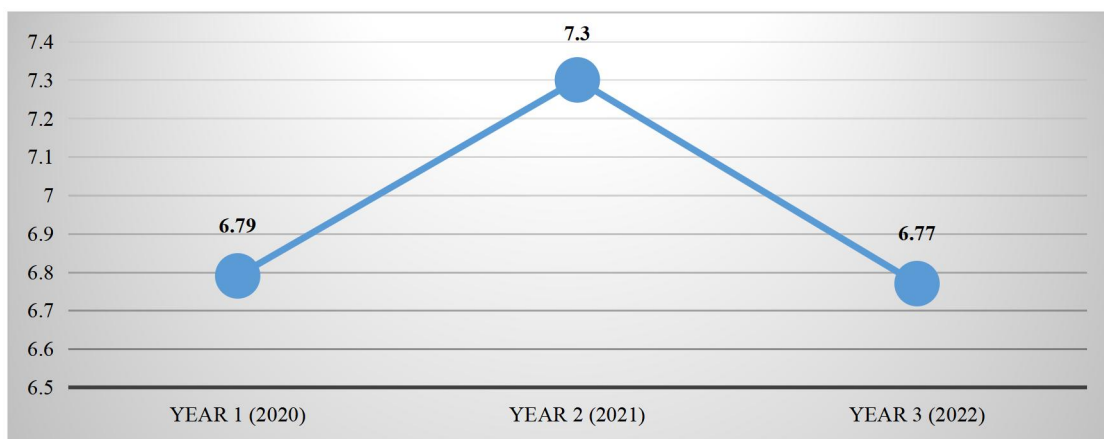


Figure 9. A graphic chart of the trend in soil (pH) analysis per year.

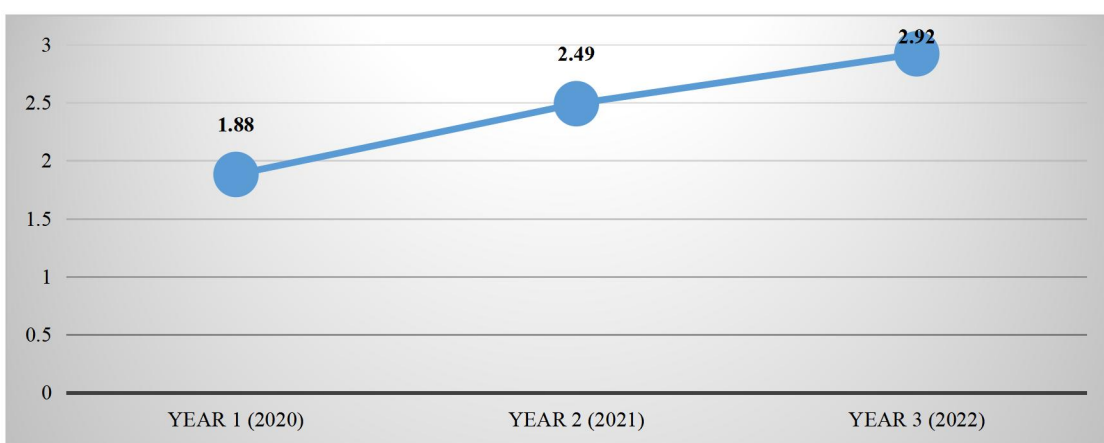


Figure 10. A graphic chart of the trend in Nitrogen Organic Matter analysis per year.

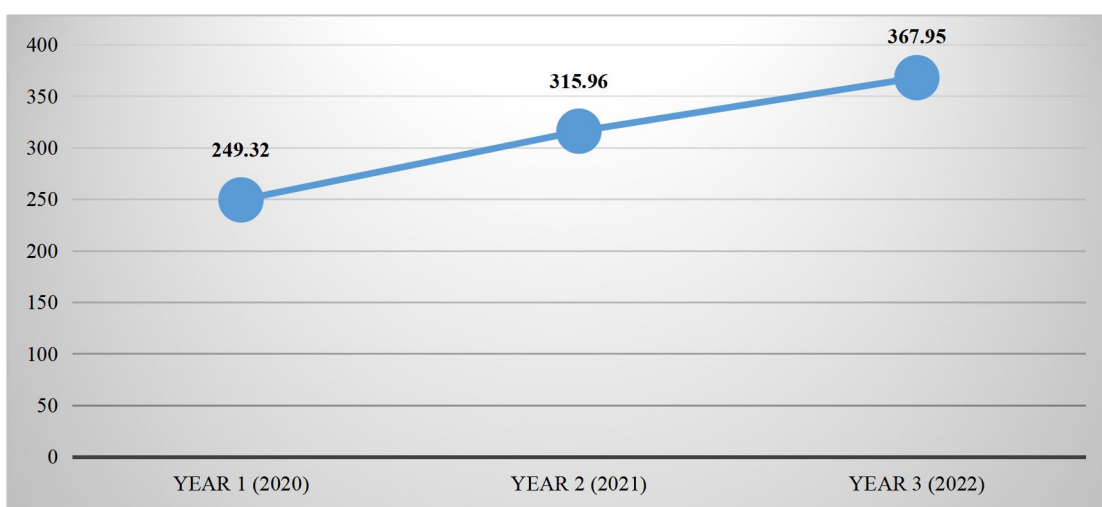


Figure 11. A graphic chart of the trend in Potassium (K) analysis per year.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 12

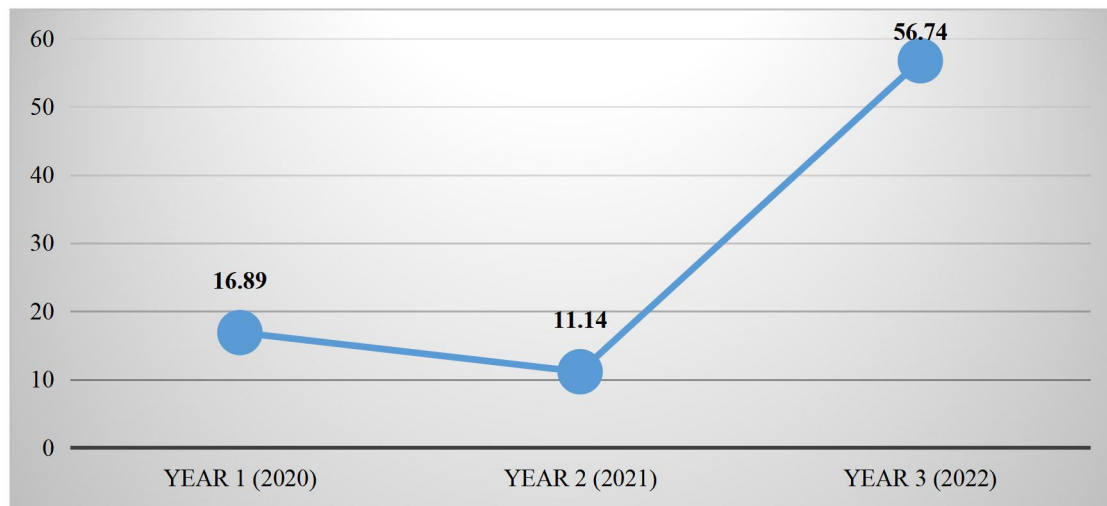


Figure 12. A graphic chart of the trend in Phosphorous (P) analysis per year.